1.

When was the C language developed?

A. 1969
B. 1972
C. 1975
D. 1980

Answer: B

2.

The assembler converts which type of code into object code (machine code)?

A. Source code
B. Preprocessed code
C. Assembly code
D. Executable code

Answer: C

3.

Which of the following statements is TRUE about an IDE (Integrated Development Environment)?

A. An IDE is only used for writing code.
B. An IDE replaces the need for a compiler.
C. An IDE provides tools like editor, debugger, compiler, all in one.
D. An IDE cannot debug programs.

Answer: C

4.

Which tool in the C toolchain helps in finding and fixing errors in a running program?

A. Compiler
B. Assembler
C. Debugger
D. Preprocessor

Answer: C

5.

The preprocessor removes the _____ from the source code.

A. comments
B. header files
C. both com
D. none of the mentioned

Answer: A

1.

What will be the output of the following C code?

```c
#include <stdio.h>
int main(void)
{
    float x = 123.45;
    int y = 120;
    printf("%6.0f\n", x);
    printf("%.4f\n", x);
    printf("%6d\n", y);
    return 0;
}
```

A.
```
   123
123.4500
   120
```

B.
```
  123.45
123.4500
120
```

C.
```
123.45
123.45
120
```

D.
```
   123.45
  123
  120
```

Answer: A

2.

Which of the following is TRUE about identifiers in C?

A. Idntifiers can contain special characters like # and @
B. Identifiers can be the same as keywords
C. Identifiers can start with an underscore or letter
D. Identifiers are case-insensitive

Answer: C

3.

Which of the following is an example of a runtime error in C Programming?

A. Lvalue error
B. object code of a function with specified name is not available
C. Divide by zero
D. Output successfully displayed but not as per problem statement requirement

Answer: C

4.

How many tokens are in the following line of code?

```c
int x = a + b * 5;
```

A. 8
B. 5
C. 6
D. 9

Answer: D

SunBeam

5.

What will be the output of the following C code?

```
#include <stdio.h>
int main(void)
{
    printf(" SunBeam\aKarad\b\t2000\rPune\n");
    return 0;
}
```

A. PuneBeamKarad 2000
B. SunBeamKarad 2000
C. PuneBeamKarad2000
D. PuneBeamKarad 2000

Answer: A

1.

Which of the following are incorrect data types in C?

A. short unsigned int
B. short float
C. long double
D. unsigned long

Answer: B

2.

What will be the output of the following c code?

```
#include<stdio.h>
int main()
{
    int a=5,b=10,c=15;

    printf("%d ",sizeof(c/=a+b));
    printf("%d",c);

    return 0;
}
```

A. 4 1
B. 4 15
C. 2 1
D. Compile time error

Answer : B

SunBeam

3.

If c is a variable of type char and i has type int, which one of the following statements is illegal ?

A. i += c
B. c = 2 * c - 1
C. putchar(c);
D. c = getchar(i);

Answer: D

4. .

Which of the following is TRUE about the sizeof operator?

A. sizeof is evaluated at runtime
B. sizeof cn be applied to functions
C. sizeof returns the number of bits
D. sizeof is a compile-time operator

Answer: D

5.

What will be the output of the following c code?

```
#include <stdio.h>
int main()
{
    char a = 200;
    unsigned char b = 200;

    if (a == b)
        printf("Sunbeam Karad\n");
    else
        printf("Sunbeam Pune\n");

    return 0;
}
```

A. Sunbeam Karad
B. Sunbeam Pune
C. Compiler Error
D. Runtime Error

Answer: B

1. What will be the output of the following C code?

```c
#include<stdio.h>
int main(void)
{
    int desd, dmc, ditiss;
    desd = dmc = ditiss = 1;
    desd += dmc += ditiss;
    printf("%d %d %d\n", desd, dmc, ditiss);

    desd -= dmc -= ditiss;
    printf("%d %d %d\n", desd, dmc, ditiss);

    desd *= dmc *= ditiss;
    printf("%d %d %d\n", desd, dmc, ditiss);
    return 0;
}
```

A. 3 2 1
   1 0 1
   1 1 1
B. 3 2 1
   2 1 1
   2 1 1
C. 3 2 1
   2 1 2
   1 1 1
D. 2 2 2
   1 0 0
   0 1 0

Answer: B

2. What will be the output of the following C code?

```c
#include<stdio.h>
int main(void)
{
    int _sunbeam = 8;

    _sunbeam = _sunbeam & ~(1 << 1);
    _sunbeam = _sunbeam ^ (1 << 1);
    printf("%d",_sunbeam);

    return 0;
}
```

A. 4
B. 8
C. 10
D. 12

Answer: C

3. What Which operator from the following has the lowest priority?

A. Assignment Operator
B. Division Operator
C. Comma Operator
D. Conditional Operator
E. Unary Operator

Answer: C

SunBeam

4.

What will be the Output of the following C code?

```c
#include <stdio.h>
int main(void)
{
    int dac = 10, dbda = 5, dmc = 5;
    int cdac;
    cdac = dac == (dbda + dmc);
    printf("%d", cdac);
    return 0;
}
```

A. Syntax error
B. 1
C. 10
D. 5

Answer: B

5.

What will be the output of the following C code?

```c
#include<stdio.h>
int main(void)
{
    int desd_pune;
    desd_pune = 0x144+0220+144;
    printf("%d",desd_pune);
    return 0;
}
```


A. 508
B. 596
C. 612
D. Runtime Error

Answer: C

SunBeam

1. What will be the output of the following c code?

```c
#include<stdio.h>
int main(void)
{
	int desd=-120 , ditiss=60*2;
	if( desd || ++ditiss)
		printf("%d",ditiss);
	return 0;
}
```

A. 120
B. -120
C. 121
D. Compile time error
E. -119

Answer: A

2. What will be the output of the following C code?

```c
#include <stdio.h>
int main(void)
{
	int karad;	karad = printf("@\b#");
	printf("%d\n", karad <=1 ? 0240 : 0120);
	return 0;
}
```

A. #160
B. @80
C. #80
D. @160
E. Compile Time Error

‘ Answer: C

3. What will be the output of the following c code?

```c
#include<stdio.h>
int main(void)
{
	float ditiss =60.0f;
	double dbda =60.0F;
	if(ditiss == dbda)
		printf("Devops");
	else
		printf("AI");
	return 0;
}
```

A. Devops
B. AI
C. Compile time error
D. Runtime error

Answer: A

4. What will be the Output of the following C code?

```c
#include<stdio.h>
int main(void)
{
	char c = 'A' ;		int cpp = 64 ;
	printf(c>cpp ? "DESD" : c==cpp+1 ? "DAC" : "DMC");
	return 0;
}
```

A. DESD
B. DAC
C. DMC
D. Compiler Error

Answer: A

5.

What will be the Output of the following C code?

```c
#include<stdio.h>
int main(void)
{
	int Sunbeam_Pune = 10;
	if (Sunbeam_Pune > 5)
	{
		Sunbeam_Pune++;
	}
	else
	{
		Sunbeam_Pune--;
	}
	if (Sunbeam_Pune < 10)
	{
		Sunbeam_Pune = Sunbeam_Pune * 2;
	}
	else
	{
		Sunbeam_Pune = Sunbeam_Pune / 2;
	}
	printf("The value of Sunbeam_Pune is = ");
	printf("%d", Sunbeam_Pune);
	return 0;
}
```

A. The value of Sunbeam_Pune is = 11

B. The value of Sunbeam_Pune is = 20

C. The value of Sunbeam_Pune is = 22

D. The value of Sunbeam_Pune is = 5

Answer: D

1.

Which of the following statements about enumeration constants are true?

A. An enum is a special type that represents a group of constants.

B. Enumeration constants have exactly the same properties as constants created using #define.

C. All constants in an enumeration must have different values.

D. None of the above

Answer : A

2.

What will be the Output of the following C code?

```c
#include<stdio.h>
enum course_capacity{DAC=240, KDAC=120, DBDA=60,
DESD=120, DITISS=60};
int main(void)
{
	typedef enum course_capacity FLOAT;
	FLOAT no_of_candidates=(FLOAT)500/200.0F;
	printf("%d", no_of_candidates);
	return 0;
}
```

A. 2.5

B. 2.0000000

C. Compiler Error

D. Garbage value

E. 2

Answer : E

3.

What will be the Output of the following C code?

```c
#include <stdio.h>
enum {PUNE, KARAD, MUMBAI, BANGLORE} cities;
int main(void)
{
    int dac = 0, desd = 0;
    cities = KARAD;     desd= KARAD;
    printf("%d\t", desd);
    switch (cities)
    {
        case PUNE: desd--;
            break;
        case KARAD: --dac;
        case MUMBAI: dac++;
            break;
        case BANGLORE: --desd;
            break;
    }
    printf("%d\n", cities);
    return 0;
}
```

A. 0 0
B. 1 1
C. 0 1
D. 1 0
E. 0 0

Answer : B

4.

What will be the Output of the following C code?

```c
#include<stdio.h>
enum courses {DAC=240 , KDAC=240/2 , DMC= KDAC ,  DESD= DMC ,
                DBDA= DMC/2, DITIS=DBDA };
int main(void)
{
    int capacity=DESD +DMC + DITIS+ +KDAC ;
    switch (capacity << 2)
    {
        default:
            printf("SunBeam DMC\n");
        case DAC:
            printf("SunBeam DAC PUNE\n");
        case KDAC:
            printf("SunBeam DAC Karad\n"); break;
    }

    return 0;
}
```

A. SunBeam DMC
   SunBeam DAC PUNE
   SunBeam DAC Karad
B. SunBeam DMC
C. SunBeam DAC PUNE
D. SunBeam DAC Karad
E. None of above

Answer : A

5.

What will be the Output of the following C code?

```
#include <stdio.h>

int main(void)
{
	int var=3;

	switch(var)
	{
		case 2+3: printf("DAC ");
		case 1+2: printf("DESD ");
		case 2*2: printf("DMC "); break;
		default: printf("DBDA ");
		case var: printf("SunBeam");break;
	}

	return 0;
}
```

A. DAC DESD DMC DBDA
B. DESD DMC DBDA
C. DESD DMC
D. DESD
E. Compile Time Error

Answer: E

1. What will be the Output of the following C code?

```
#include<stdio.h>
int main(void)
{
	int dac=4,kdac;
	for(kdac=dac-1; dac>0, kdac>0;--dac,kdac=dac-1)
		printf("%d ", dac- kdac);
	return 0;
}
```

A. 4 2 0
B. 1 2 3
C. 3 2 1
D. 1 1 1
E. Compile time error

Answer: D

2. What will be the Output of the following C code?

```
#include <stdio.h>
int main(void)
{
	int value=4,num=0;
	while(num<=~(~value))
		num++;
	printf("%d ",num);	return 0;
}
```

A. 4
B. 5
C. 0
D. 1
E. infinite loop

Answer: B

3.

What will be the Output of the following C code?

```
#include<stdio.h>
int main(void)
{
    int course_no=0,no_students = 0, total_courses=5;

    for (;course_no<(total_courses*2);course_no++)
    {
        if (course_no % 2)
            continue;

        no_students += course_no;
    }

    printf("%d\n", no_students);

    return 0;
}
```

A. 10

B. 20

C. 30

D. 40

E. infinte loop

Answer: B

4.

What will be the Output of the following C code?

```
#include <stdio.h>
int main(void)
{
    int sunbeam_karad = 10, sunbeam_pune=2;

    while (sunbeam_karad)
    {
        printf("%d ", sunbeam_karad++);

        sunbeam_karad /=sunbeam_pune ;
    }

    return 0;

}
```

A. it prints 10 8 6 4 2 and then 1 infinite times

B. it prints 10 5 3 2 and then 1 infinite times

C. it prints 10 infinite times

D. it prints 10 9 8 7 6 5 4 3 2 1

E. compile time erorr

Answer: B

5.

What will be the Output of the following C code?

```c
#include <stdio.h>
int main(void)
{
	int course = 0, no_of_courses=5, no_of_centers=2;

	start:
	if (course < (no_of_courses-no_of_centers))
	{
		printf("%d ", course);
		course++;
		goto start;
	}

	return 0;
}
```

A. 0 1 2
B. 0 1 2 3
C. 1 2 3
D. Infinite Loop
E. Compile time error

Answer: A

1.

What will be the Output of the following C code?

```c
#include <stdio.h>
int sunbeam_res(int dac_pass,int dmc_pass,int dbda_pass)
{
	if ((dac_pass >= 0 && dac_pass < dbda_pass - 1) &&
	  (dmc_pass >= 0 && dmc_pass < dbda_pass - 1))
		return 1;
	return 0;
}
int main(void)
{
	int desd=5 , dmc=10, dac=desd+dmc;
	printf("%d and %d:%d\n",desd,dmc,
		sunbeam_res(desd,dmc,dac));
	return 0;
}
```

A. 5 and 10: 1
B. 5 and 10: 2
C. 5 and 10: 0
D. Compile Time Error

Answer: A

2.

Which of the following would be valid prototypes for a f unction that returns nothing and has one double parameter?

A. void f(double x)
B. void f(double);
C. void f(x);
D. Both A and B

Answer:D

3.

Which storage class is used primarily to indicate that a variable or function can be shared in multiple files?

A. static
B. global
C. extern
D. register

Answer: C

4.

Consider the following code

```
#include<stdio.h>
extern int s;
int t;
static int u;
int main(void)
{
	return 0;
}
extern int t;
```

Which variables out of s, t and u can be available to a function present in all files of the project? Assume variables are defined in either of one file.

A. Only u
B. s & u
C. s, t, u
D. t & s

Answer :D

5.

Which of the following declarations are Illegal?

[Consider PI= 3.142]

A. `char c = 65;`
B. `static int i = 5, j = i * i;`
C. `double d = 2 * PI;`
D. `double angles = (0, PI / 2, PI, 3 * PI / 2);`

Answer: B

SunBeam

1.

What will be the Output of the following C code?

```c
#include <stdio.h>
void print(int n);
int main(void)
{
	int value = 16;

	print(value);

	return 0;
}
void print(int Value)
{
	if (Value)
	{
		print(Value/2);
		putchar('0' + Value % 2);
	}

	return;
}
```

A. 10101

B. 00001

C. 10001

D. 10000

E. Garbage Values

Answer: D

2.

What will be the Output of the following C code?

```c
#include <stdio.h>
int Value = 6;
int main(void)
{
	static int value = 9;
	printf("%3d",value-=3);
	if(value)
		main();
	return 0;
}
```

A. 9 6 3 0

B. 6 3 0

C. stack overflow

D. 9 6 3

E. 6 3

Answer: B

3. What will be the Output of the following C code?

```c
#include<stdio.h>
int course_count(int count)
{
	return(count ? 1+course_count(count & count -1):0);
}
int main(void)
{
	static int dac, kdac, desd, dmc, dbda, ditis;
	printf("%d",
	course_count((dac+kdac+desd+dbda+ditis+6)));
	return 0;
}
```

A. 0
B. 1
C. 2
D. Compiler Error

Answer: C

4.

What will be the Output of the following C code?

```
#include<stdio.h>
int Value;
void increment_rec(int value)
{
	if(value<=3)
		printf("%d",value);
	increment_rec(value++);
}
int main(void)
{
	int value=Value+1;
	increment_rec(value);
	return 0;
}
```

A. 3 2 1
B. 1 2 3
C. 1 1 1
D. Runtime Error: Stack Overflow
E. Compile time eroror

Answer: D

5.

What will be the Output of the following C code?

```
#include<stdio.h>
int DAC=240, KDAC=120;
int sunbeam_diff( int dac, int kdac)
{	if (kdac== 0)
		return kdac;
	else
		return (dac * sunbeam_diff(dac, kdac - 1));
}
int main(void)
{	int result= sunbeam_diff(1,3);
	printf("%d", result);
	return 0;
}
```

A. Output will be 0 always if dac and kdac be any value
B. Output will always be b, if dac and kdac be any value
C. Computing dac * kdac
D. Computing dac + kdac
E. garbage values

Answer: A

1.

What will be the Output of the following C code?

```c
#include<stdio.h>
int value1=1,value2=2;
void change_values(int *ptrvalue1,int *ptrvalue2)
{
	ptrvalue2=ptrvalue1;
	*ptrvalue2=3;
	return;
}
int main(void)
{
	const int Value1=100, Value2=200;

	change_values(&value1,&value1);

	printf("value1=%d value2=%d\n",value1,Value2);

	return 0;
}
```

A. value1=3 value2=2
B. value1=3 value2=1
C. value1=2 value2=3
D. value1=2 value2=200
E. value1=3 value2=200

Answer: E

2.

If ptr_karad is an integer pointer which stores address 1024 from memory,then what will be the value of ptr_karad + 5 ?

```c
#include<stdio.h>
int main(void)
{
	int const *ptr_karad=(int const*)1024;

	printf("\n ptr_karad=%u %u",ptr_karad, ptr_karad+5);
	return 0;
}
```

A. ptr_karad=1024 1044
B. ptr_karad=1024 1029
C. ptr_karad=1024 1064
D. ptr_karad=1024 1034
E. compile time error
F. ptr_karad=1044 1044

Answer: A

3.

If sunbeam is a variable and ptr_sunbeam points to sunbeam, which of the following expressions are aliases for sunbeam?

A. *ptr_sunbeam
B. &ptr_sunbeam
C. *&sunbeam
D. &*ptr_sunbeam
E. both A and C

Answer: E

4.

Suppose that ptr_sunbeam_pune has been declared as follows what will be the legal statements?

```
#include<stdio.h>
int main(void)
{
    const char sunbeam_pune[]="Hinjewadi";
    const char *const ptr_sunbeam_pune=sunbeam_pune;
    return 0;
}
```

A. putchar(ptr_sunbeam_pune);

B. putchar(*ptr_sunbeam_pune);

C. puts(ptr_sunbeam_pune);

D. puts(*ptr_sunbeam_pune);

E. both B & C

Answer : E

5

```
#include<stdio.h>
int main(void)
{
    char data='A';  const char *ptr=NULL;   ptr=&data;
    printf("%d %d %d",sizeof(*ptr),sizeof(ptr),sizeof('A'));
    return 0;
}
```

What will be the Output of the above C code?

A. 4 4 4

B. 1 1 1

C. 1 4 4

D. 1 8 8

E. 8 8 8

Answer: C

1. What will be the Output of the following C code?

```
#include<stdio.h>
int main(void)
{
    int sunbeam[7]={11,22,33,44,55};
    int * const ptr_sunbeam=&sunbeam[6];
    printf("%d ",(*ptr_sunbeam+=5) + sunbeam[4]);
    printf("%d",*ptr_sunbeam + sunbeam[5]);
    return 0;
}
```

A. 60 60

B. 60 5

C. 5 5

D. 55 5

E. garbage values

F. Compile time error

Answer: B

2. What will be the Output of the following C code?

```
#include<stdio.h>
int main( void )
{   int array_capacity[6]={240, 120 , 120 , 120 ,60, 60};
    printf("%d",0[array_capacity] + -4[array_capacity]);
    return 0;
}
```

A. 180

B. 300

C. Compiler error

D. -180

E. garbage values

Answer: A

3.

What will be the Output of the following C code?

```c
#include<stdio.h>
int main(void)
{
	char strMessage[]="SunBeam Pune & Karad";
	char *ptr_Message=NULL;
	ptr_Message=strMessage;
	ptr_Message+=3;
	strMessage++;
	printf("%s",ptr_Message);
	return 0;
}
```

A. Beam Pune & Karad

B. eam Pune & Karad

C. nBeam Pune & Karad

D. Compile time Error

E. garbage value

Answer : D

4. Suppose that sunbeam is a one-dimensional array of int and ptr_sunbeam is a pointer variable of int type.
If we store address of 0th element of array sunbeam &sunbeam[0] in to pointer sunbeam, which of the following expressions will not give true result ?

A. ptr_sunbeam == &sunbeam[0]

B. ptr_sunbeam == sunbeam[0]

C. *ptr_sunbeam == sunbeam[0]

D. ptr_sunbeam[0] == sunbeam[0]

E. &*ptr_sunbeam == &sunbeam[0]

Answer: B

5.

What will be the Output of the following C code?

```c
#include<stdio.h>
int main(void )
{
	int sunbeam[] = {60, 120, 180, 144, 90, 240};
	int *ptr_dac=&sunbeam[1],*ptr_kdac = &sunbeam[5];
	int dac,kdac;
	dac=ptr_dac<ptr_kdac?240:120;
	kdac=*ptr_dac<*ptr_kdac?120:240;
	printf("ptr_dac=%d,ptr_kdac=%d",dac,kdac);
	return 0;
}
```

A. ptr_dac=120,ptr_kdac=120

B. ptr_dac=240,ptr_kdac=240

C. ptr_dac=240,ptr_kdac=120

D. ptr_dac=garbage value,ptr_kdac=garbage value

Answer: C

1.

What will be the Output of the following C code?

```
#include <stdio.h>
int main(void)
{
	char sunbeam_course[]="PG-DITTIS";
	char *ptr_course=NULL;
	ptr_course=sunbeam_course;
	for (    ; *ptr_course ;  ptr_course++ )
		--*ptr_course;
	puts(sunbeam_course);	printf("\n");
	return 0;
}
```

A. FO , DITTIS
B. FO , EJUUJT
C. OF , CHSSHR
D. OF , RHSSHC
E. garbage values

Answer: C

2.

Choose the function that is most appropriate for reading in a multi-word string?

A. strnset()
B. scanf("%[^.]s,stribg_name); // scan up to .
C. strchr()
D. gets()
E. Both B and D

Answer: E

3.

What will be the Output of the following C code?

```
#include<stdio.h>
#include<string.h>
int main(void)
{
	char sunbeam_course[20];
	strcpy(sunbeam_course,"PG-DESD\0PG-DBDA\0");
	printf("%d ", strlen(sunbeam_course));
	printf("%d ",sizeof(sunbeam_course));
	printf("%s ", sunbeam_course+3);
	return 0;
}
```

A. 7 20 DESD
B. 15 20 PG-DESD
C. 20 20 PG-DESD PG-DBDA
D. 20 7 PG-DESD
E. 20 15 PG-DESD PG-DBDA

Answer: A

4.

What will be the Output of the following C code?

```
#include<stdio.h>
#include<string.h>
int main(void)
{
	printf("%c %c %c ", "KDAC"[0], "DESD"[3] ,"DAC"[1] );
	printf("%c[ %d ]","DMC"[2] ,"DBDA"[1]+"DITTIS"[2]-30);
	return 0;
}
```


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A. K A C D [120]
B. K D A C [150]
C. K D A C [120]
D. Compile time error
E. K C A D [120]

Answer: C

5.

What will be the Output of the following C code?

```
#include<stdio.h>
int main(void)
{
    char s[] = "SUNBEAM";   int i=0;    char ch;
    ch = s[i++];
    printf("%c", ch+32);
    ch = s[++i];
    printf("%c", ch+' ');
    ch = ++i[s];
    printf("%c", ch +' ');
    ch = i++[s];
    printf("%c", ch +32);
    return 0;
}
```

A. SNOO
B. SOON
C. snoo
D. SONO
E. sono

Answer: C

1.

What will be the Output of the following C code?

```
#include<stdio.h>
#define NO 6
#define LEN 8
int main(void)
{
    char courses[NO][LEN]={"PG-DAC","PG-DESD","PG-DMC", "PG-DBDA","Pre-Cat"};
    printf("%d %d / ",sizeof(courses),sizeof(courses[LEN]));
    printf("%d %d",sizeof(courses[NO-1][LEN-1]) , sizeof(**courses));
    return 0;
}
```

A. 48 4 / 1 1
B. 48 8 / 1 1
C. 48 4 / 1 4
D. 48 4 / 1 8
E. 48 4 / 1 1

Answer : B

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2.

What will be the Output of the following C code?

```c
#include<stdio.h>
#define ROW 5
#define COL 8
int main(void)
{
	static char arr[ROW][COL]={"PRECAT","PG-DAC","PG-DMC","PG-DBDA","PG-DESD"};
	printf("%c, %s", *(arr[3]+3), (*(arr+3)+3));
	return 0;
}
```

A. D, DBDA
B. P, PG-DBDA
C. D, PG-DMC
D. D, DMC
E. D, garbage value

Answer: A

3.

What will be the Output of the following C code?

```c
#include<stdio.h>
int main (void)
{
	static char *loc[]={"sunbeam","SunBeam","sUNbEAM", "SUNBEAM" };
	char **ptrLoc[]={loc+3, loc+2, loc+1, loc}, ***ptr_add_loc = NULL;
	ptr_add_loc = ptrLoc;
	**++ptr_add_loc;
	printf("%s",**++ptr_add_loc + 3);
	return 0;
}
```

A. BEAM
B. Beam
C. bEAM
D. beam
E. compile time error

Answer: B

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4.

What will be the Output of the following C code?

```c
#include<stdio.h>
#define NO 6
#define LEN 9
int main(void)
{
	char courses[NO][LEN]={"PG-DAC","PG-DESD","PG-DMC","PG-DBDA","PreCat"};
	char *ptr=(char *)courses[0];
	printf(" %s", (ptr+3)+*(ptr+1)-ptr[1]);
	printf("-%c",*(ptr+3)+*(ptr+1)-ptr[1]);
	return 0;
}
```

A. PG-DAC P

B. Compiler Error

C. DAC-D

D. NULL-0

E. garbage values

Answer: C

5.

What will be the Output of the following C code?

```c
#include<stdio.h>
#include<string.h>
int main()
{
	static char names[5][20]={"PG-DAC","PG-DMC","PG-DBDA","PG-DITIIS"};
	int i;
	char t[20];
	for (i=0;i<=3;i++)
		printf(" %s ",names[i]);

	strcpy(t,names[2]);
	strcpy(names[2],names[3]);
	strcpy(names[3],t);
	printf("\n");

	for (i=0;i<=3;i++)
		printf(" %s ",names[i]);
	return 0;
}
```

A. PG-DAC PG-DMC PG-DBDA PG-DITIIS
PG-DAC PG-DMC PG-DITIIS PG-DBDA
B. PG-DAC PG-DMC PG-DBDA PG-DITIIS
PG-DAC PG-DMC PG-DBDA PG-DITIIS
C. PG-DAC PG-DMC PG-DITIIS PG-DBDA
PG-DAC PG-DMC PG-DBDA PG-DITIIS
D. Garbage values
E. Compile time error

Answer: A

1.

If following program having name cmdline is run from the command line as
".\cmdline.exe PG-DAC PG-DMC PG-DBDA"

```
#include <stdio.h>
#include <string.h>
int main(int argc, char *argv[])
{
    char **ptr_course=NULL;
    for (ptr_course = argv + argc - 1; ptr_course > argv; ptr_course--)
        printf("%s ", *ptr_course);
    return 0;
}
```

A. PG-DAC PG-DMC PG-DBDA
B. PG-DBDA PG-DMC PG-DAC
C. No Output
D. Compiler Error

Answer: B

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2. If following program having name cmdline is run from the command line as
".\cmdline.exe Sunbeam Hinjawadi 600 Karad 120"

```
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char* argv[])
{
	int sum = 0, counter=0;
	char **ptr=NULL;
	for (ptr = argv ; ptr < argv + argc; ptr++)
		sum += atoi(*ptr), counter++;
	printf("Total: %d [%d]", sum, counter);
	return 0;
}
```

A. Total: 600 [5]
B. Total: 720 [5]
C. Total: 720 [6]
D. Compiler time error
E. Total: 120 [6]

Answer : C

3. If following program having name cmdline is run from the command line as
".\cmdline.exe Pre-CAT PG-DBDA PG-DAC"

```
#include<stdio.h>
int main(int argc, char *argv[],char *envp[])
{
	argc--;
	while(argc)
	{
		printf("%c ", argv[argc--][4] + ' ' );
	}
	return 0;
}
```

A. A B C
B. a b c
C. C B A
D. Compile time error
E. c b a

Answer: B

4.

If following program having name cmdline is run from the command line as "cmdline" then what would be the output?

```
#include<stdio.h>
int main(int argc, char *argv[], char *envp[])
{
	int index;
	for(index=1; envp[index]!=NULL; index++)
		printf("\n%d] %s",index, envp[index]);

	return 0;
}
```

A. List of all environment variables
B. List of all command-line arguments
C. count of command-line arguments
D. Error: cannot have more than two arguments in main()
E. compile time error

Answer: A

5.

If the following program having the name cmdline is run from the command line as:

```
#include <stdio.h>
int main(int argc, char *argv[])
{
	puts(argv[0]);
	return 0;
}
```

What will be the output?

A. prints the name of the source code file.
B. prints argv.
C. prints the number of command line arguments.
D. prints the name of the executable code file.

Answer: D

1.

What will be the Output of the following C code?

```c
#include<stdio.h>
#define DAC_PUNE 120 + 120
#define DAC_KARAD 0 + 120
int main()
{

    printf("Total DAC students at SunBeam = ");
    printf("%d\n ", DAC_PUNE * DAC_KARAD + DAC_KARAD);
    return 0;
}
```

A. Total DAC students at SunBeam =240
B. Total DAC students at SunBeam =360
C. Total DAC students at SunBeam =28920
D. Compile time Error
E. Total DAC students at SunBeam =garbage value

Answer: B

2.

What will be the output of the following c code?

```c
#include<stdio.h>
#define NO_OF_STUDENTS 20
#define DAC_SUNBEAM(dac_sunbeam_group_count)  dac_sunbeam_group_count*NO_OF_STUDENTS
int main(void)
{
    #undef NO_OF_STUDENTS
    #define NO_OF_STUDENTS 30
    int dac_sunbeam_group_count=12, value=0;
    value= DAC_SUNBEAM(dac_sunbeam_group_count);
    printf(" Total DAC Students = %d",value);
    return 0;
}
```

A. compile time error
B. Total DAC Students = garbage value
C. Total DAC Students = 360
D. Total DAC Students = 240
E. Run time error

Answer: C

3.

What will be the output of the following c code?

```c
#include <stdio.h>
#define dac_karad 240
int main(void)
{
	#define dac_karad 120

	printf("DAC_Karad Count=%d",dac_karad);

	return 0;
}
```

A. DAC_Karad Count=240
B. DAC_Karad Count=120
C. Runtime error
D. Compile time error
E. redefining error for dac_karad

Answer:B

4.

What will be the output of the following c code?

```c
#include<stdio.h>
#define no_of_course 5
#define DIM( array, type) sizeof(array)/sizeof(type)
int main(void)
{
	int courses[ no_of_course * 2 ];
	printf("The dimension of the array is %d", DIM(courses, int));
	return 0;
}
```

A. The dimension of the array is 2
B. The dimension of the array is 20
C. The dimension of the array is 10
D. Compile time Error
E. The dimension of the array is garbage value

Answer: C

5.

Which directive is used to undefined a previously defined macro?

A. #ifdef
B. #ifndef
C. #undef
D. #endif
E. #undefined

Answer: C

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1.

What will be the Output of the following C code?

```
#include<stdio.h>
#include<stdlib.h>
int main(void)
{
    int *ptrkaraddac = (int*)malloc(sizeof(int));    *ptrkaraddac=120;
    int *ptrpunedac = ptrkaraddac;
    *ptrpunedac= * ptrkaraddac + *ptrkaraddac ;
    printf("\n %d",*ptrpunedac / *ptrkaraddac );
    free(ptrkaraddac);
    ptrkaraddac=ptrpunedac=NULL;
    return 0;
}
```

A. 1

B. 2

C. Compile time error

D. Run time erorr

E. garbage values

Answer: A

2. The given below program allocates the memory, what function will you use to free the allocated memory?

```
#include<stdio.h>
#include<stdlib.h>
#define MAXROW 4
#define MAXCOL 5
int main (void)
{
    int **ptrMat=NULL;
    ptrMat = (int **) malloc(MAXROW * sizeof(int*));
    return 0;
}
```

A. memfree(int ptrMat);

B. free(ptrMat);

C. dealloc(ptrMat);

D. Both, free(ptrMat); and dealloc(ptrMat);

E. none of above

Answer: B

3.

What is your comment on the below C statement?

```
signed int *ptrValue=(int*)malloc(sizeof(unsigned int));
```

A. Improper type casting
B. Would throw Runtime error
C. Memory will be allocated but cannot hold an int value in the memory
D. No issue with statement

Answer: D

4.

Which of the following statements is true about malloc/calloc ?

A. After a call of a malloc/calloc it returns the value of the memory block which can be used to store data.
B. Return type of function is void pointer which can not be converted to required type
C. After call of function if memory is not available function returns NULL
D. Memory is given from stack section

Answer: C

5.

What will be the Output of the following C code?

```
#include <stdio.h>
#include<stdlib.h> int* set()
{
	static int *to=NULL;

	to = (int*)malloc(5 * sizeof(int));
	return to;
}
void solve()
{
	int *ptr=NULL, i=1;
	ptr=set();
	for(*ptr = 10; i<=5; i++)
		printf(" %d \t", *(ptr+i)= *ptr+i );
	free(ptr);
	return
}
```

```
int main(void)
{
    solve();
    printf("\n");
    return 0;
}
```

A. 10 10 10 10 10
B. Garbage values
C. 11 12 13 14 15
D. 10 20 30 40 50
E. Run time error

Answer: C

1.

what will be the Output of the following C code?

```
#include <stdio.h>
#pragma pack(1)
struct
{
    double dac;
    union
    {
        char desd[4];
        double dbda;
        int dittis;
    }karad;
    char dmc[4];
}sunbeam;

int main(void)
{
    printf("%d",sizeof(sunbeam)/sizeof(sunbeam.karad));

    return 0;
}
```

A. 2
B. 3
C. 2.500000
D. 2.50
E. Compile time error

Answer: A

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2.

Given the following declarations:

```
#include <stdio.h>
struct
{
	union
	{
		char desd, dmc;
		int dbda;
	} pune;
	int dittis[5];
} karad, *ptrLocarion = &karad;

int main(void)
{
	return 0;
}
```

Which of the following statements are valid?

A. ptrLocarion->dmc = ' ';
B. ptrLocarion->dittis[3] = 60;
C. (*ptrLocarion).pune.desd = '*';
D. ptrLocarion->pune->dbda = 20;
E. Both B and C

Answer : E

3.

what will be the Output of the following C code?

```
#include <stdio.h>
union
{
	float value;
	struct float_value
	{
		unsigned int fraction: 23;
		unsigned int exponent: 8;
		unsigned int sign: 1;
	} m;
}f;
int main(void)
{
	f.m.sign = 1;
	f.m.exponent = 128;
	f.m.fraction = 0;
	printf("%.1f\n", f.value);
	return 0;
}
```

A. 0.0
B. -1.0
C. -2.0
D. -3.0
E. garbage value

Answer: C

4.

what will be size of struct tree node?

[Note: consider 64 bit compiler]

```
#include<stdio.h>
#pragma pack(1)
struct tree
{
	struct test *left;
	double data;
	struct test *right;
};
int main(void)
{

	printf("\n size of node=%d", sizeof(struct tree));

	return 0;
}
```

A. 16
B. 12
C. 20
D. 24
E. compile time error

Answer: D

5. what will be the Output of the following C code?

```
#include <stdio.h>
int main(void)
{
	struct location
	{
		double karad;
		double pune;
	}l1 = {0.0, 1.0}, l2 = {1.0, 0.0}, l3;
	l1 = l2;
	l3.karad = l1.karad+ l2.karad;
	l3.pune = l1.pune + l2.pune;
	printf("l1:karad=%.1f,",l1.karad);
	printf("pune=%.1f \n",l1.pune);
	printf("l2:karad=%.1f,",l2.karad);
	printf("pune=%.1f \n",l2.pune);
	printf("l3:karad=%.1f,",l3.karad);
	printf("pune=%.1f \n",l3.pune);
	return 0;
}
```

A. l1: karad = 1.0, pune = 0.0
   l2: karad = 1.0, pune = 0.0
   l3: karad = 2.0, pune = 0.0
B. l1: karad = 0.0, pune = 1.0
   l2: karad = 1.0, pune = 0.0
   l3: karad = 1.0, pune = 1.0
C. l1: karad = 1.0, pune = 1.0
   l2: karad = 1.0, pune = 1.0
   l3: karad = 1.0, pune = 1.0
D. Compiler time error

Answer: A

1.

if file contents following data in SunBeam.txt
[PreCAT PG-DAC PG-DESD PG-DMC PG-DBDA PG-DITISS]

```
#include <stdio.h>
#include <stdlib.h>
int main(void)
{
	FILE *fptr=fopen("SunBeam.txt","r");
	if( fptr!= NULL)
	{
		char ch='P', src[]="G-DESD course";
		printf("%c%s", ch, src);
		while(feof(fptr))
		{
			ch=fgetc(fptr);
			printf("%c",ch);
			fseek(fptr,-1,SEEK_CUR);
		}
		fclose(fptr);
		printf(" Started at SunBeam Pune\n");
	}
	exit(0);
	return 0;
}
```

A. P
B. Infinite loop
C. No output
D. PG-DESD course Started at SunBeam Pune
E. r

Answer: D

2.

If you want to rename any file programmatically, which function you can use ?

A. int rename(const char *old, const char *new);
B. int rename(const char *new, const char *old);
C. int file_rename(const char *old, const char *new);
D. int rename_file(const char *new, const char *old);

Answer: A

3.

return type of fread and fwrite functions?

A. size_t
B. int
C. void*
D. void

Answer: A

4.

A mode which is used to open an existing binary file for both reading and writing

A. "wb"
B. "wb+"
C. "r+"
D. "rb"
E. "ab"

Answer: B

5.

Identify the invalid constant used in the fseek() function as whence' reference.

A. SEEK_SET
B. SEEK_CUR
C. SEEK_BEG
D- SEEK_END

Answer: C

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